From Supply and Use Tables to Reproducible Social Accounting Matrices: A Transparent Construction and Audit Framework

Evidence from three publication regimes: South Africa, Kenya, and the Netherlands — working paper, draft v0.8

Kodzovi Abalo*

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Abstract

Social accounting matrices (SAMs) underpin much applied policy modelling, yet the way they are built has a reproducibility problem: heterogeneous official tables are reconciled through undocumented spreadsheet operations and expert judgments that cannot be audited, updated, or transferred. This paper demonstrates that the problem is unnecessary—that SAM construction can move from an artisanal exercise to an auditable statistical-production process. We introduce a construction and audit framework in which every source is fingerprinted, every classification and formula is machine-readable, every balancing adjustment is logged at the cell where it acts, and accounting identities are enforced as executable checks. We apply it to build a complete 2019 SAM for South Africa from publicly obtainable official sources and access-controlled survey microdata, audited cell by cell against the published UNU-WIDER benchmark of van Seventer and Davies (2023): the benchmark's production and commodity structure is reproduced exactly, and its institutional macro structure is reproduced exactly except for one unpublished input (separately identified and priced at +16.1 per cent on import duties under its public proxy) and one R596 million discrepancy consistent with an unrecorded reconciling adjustment. Where the benchmark's choices are undocumented, the framework recovers them numerically; every reported discrepancy was first detected by deterministic reconciliation and then interpreted against the source record. Two further, deliberately shallower applications test the framework at the ends of the public-data spectrum: in Kenya, auditing proceeds exactly as far as the boundary set by publication format; in the Netherlands, the framework generates and balances a macro SAM from three Eurostat API calls with no benchmark consulted. The open toolkit that implements the framework is offered as reproducible infrastructure, and the central proposal follows: cell-level provenance and machine-readable construction rules, published alongside every SAM, would make such audits—and such SAMs—routine.

Keywords: social accounting matrix, supply and use tables, reproducibility, data provenance, statistical auditing, South Africa, Kenya, Netherlands.
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1 Introduction

Social accounting matrices organise an economy's transactions—production, factor incomes, institutional transfers, accumulation, and external flows—into a single square array in which every account's receipts equal its payments. They are the standard database for multiplier analysis and computable general equilibrium modelling, and in many countries they are the principal quantitative bridge between official statistics and policy simulation.

Yet SAM construction has a reproducibility problem. The way matrices are built has changed remarkably little since Pyatt and Round (1985): heterogeneous official tables are reconciled through long chains of spreadsheet operations, expert judgments, and balancing passes that are rarely documented at the level where they act—the individual cell. The consequences are familiar to every practitioner: published SAMs are trusted and used, but they can seldom be reproduced, audited, or updated by anyone other than their authors; knowledge of how a matrix was actually built leaves when its builder does; and each new country or benchmark year restarts the work nearly from scratch. Section 2 argues that this is a property of the prevailing *medium*—prose describing spreadsheets—rather than of any particular team.

This paper demonstrates that the problem is unnecessary: SAM construction can move from an opaque artisanal exercise to an auditable, executable statistical-production process. We introduce a construction and audit framework with four commitments: every source file is fingerprinted and registered; every classification, concordance, and cell formula is machine-readable and reviewable; every balancing adjustment is logged at the cell where it acts; and the accounting identities that define a SAM are enforced as executable checks rather than assumed. We then apply the framework end to end, building a complete 2019 SAM for South Africa from publicly obtainable official sources and access-controlled survey microdata—inputs available to any researcher at no cost, with no privileged access[1]—and auditing it, cell by cell, against the strongest available benchmark: the 2019 South African SAM published by UNU-WIDER (van Seventer and Davies, 2023), which is unusually well documented by the standards of the genre and ships with a technical note mapping its macro structure to South African Reserve Bank (SARB) national-accounts series.

South Africa is the evidence, not the point. The audit produces three kinds of result that we argue generalise. First, *exact replication where the sources allow it*: the benchmark's entire production and commodity side reproduces from the official supply and use tables to within R0.5 million on all 61 activities and all commodities, and 43 of 45 computable institutional macro cells match the SARB series exactly (Table 2 summarises). Second, *numerical recovery of undocumented choices*: where the benchmark's documentation is silent or mistaken, the data identify what was actually done—the edition and measure behind its capital-stock split, the unique sign structure of a misprinted aggregation formula, and the household grouping variable. Third, *a priced replicability boundary*: the benchmark rests on exactly one non-public input, and substituting its public proxy changes import duties by +16.1 per cent and product taxes by

[1]"Publicly obtainable" means: published statistical releases plus survey microdata obtainable by any registered researcher under DataFirst's standard access conditions. The one input that is not obtainable at all—the benchmark's unpublished customs figure—is treated separately and priced (Section 5.2); the *public-proxy* scenario of our SAM uses published sources only.

−1.5 per cent; a small unrecorded balancing adjustment and a handful of sector-level share edits are likewise isolated and measured (each detected because it left an arithmetic trace, which is the limit of what this method can claim).

Two further applications then place South Africa on a spectrum, and the three-country design is deliberately asymmetric: one deep audit, one data-boundary test, one transfer-and-generation demonstration. The applications are not three comparable country studies—each answers a different question about the same framework under a different publication regime, and their unequal depth is the research design, not uneven execution. Applied to Kenya—where the machine-readable supply–use detail behind IFPRI's widely used 2019 SAM is not published—the framework verifies the matrix's structure and macro controls against the national accounts and locates exactly where verification must stop: the published accounts carry a statistical discrepancy of 3.6 per cent of GDP, a balanced SAM cannot carry one, and the deviation pattern is consistent with the discrepancy being absorbed primarily through the consumption and trade aggregates. Applied to the Netherlands—where harmonised tables are a public API—the framework needs no benchmark at all: three API calls, a 70-line reader, and one script generate a 2019 macro accounting matrix whose inputs pass every identity exactly and whose residuals are each attributed to a named ESA boundary item, and a logged balancing step then closes it into a macro SAM (Section 6).

Our contribution is therefore not a new balancing estimator. That literature is active rather than closed—cross-entropy estimation (Robinson et al., 2001), GRAS for matrices with negative entries (Junius and Oosterhaven, 2003; Temurshoev et al., 2013), and KRAS for conflicting and unequally reliable constraints (Lenzen et al., 2009) each address a different compilation problem—and this paper adds nothing to it. What we contribute instead is a demonstration, on a full-scale national SAM, of five things at once: a reproducible construction pipeline; an audit methodology whose discrepancies are detected by *deterministic reconciliation against accounting identities* rather than by reading documentation; a reconstruction of a published national SAM from publicly obtainable sources with every departure disclosed and quantified; a taxonomy of the ways documentation and data diverge in even the best-documented matrices; and an open toolkit—readers for supply–use, SAM, central-bank, and survey sources; concordance and account-taxonomy machinery; RAS balancing with mandatory diagnostics; a generator for workbooks whose totals and balance checks are live formulas—offered as reproducible infrastructure for official SAM construction and released with the replication package.[2] The framework doubles as a practical construction workflow: the pipeline generates and balances the matrix mechanically, so that scarce expert time is spent only on the residual cells that identities cannot determine (Section 7.2).

The idea the paper keeps returning to is simple: *cell-level provenance and machine-readable construction rules, published alongside every SAM, would make such audits—and such SAMs—routine.* The paper proceeds as follows. Section 2 situates the work and contrasts the prevailing construction workflow with the framework's. Section 3 describes the source registry. Section 4 sets out the pipeline and the audit method. Section 5 reports the South African replication block

[2] The toolkit (`edikit`) and all scripts, registries, and derived tables are part of the replication package; see the Reproducibility and AI disclosure section.

by block. Section 6 presents the Kenyan and Dutch applications and the replicability spectrum they trace. Section 7 draws out the taxonomy, the construction workflow, and implications for statistical practice, and Section 8 concludes.

2 Background: the practice and its reproducibility problem

2.1 Related work

The SAM as an integrated accounting framework descends from Stone (1962) and was consolidated as a planning tool by Pyatt and Thorbecke (1976) and Pyatt and Round (1985). The mechanics of reconciling inconsistent accounts are equally mature: biproportional (RAS) adjustment and its generalisations—GRAS for matrices carrying negative entries (Junius and Oosterhaven, 2003; Temurshoev et al., 2013), KRAS where constraints conflict or differ in reliability (Lenzen et al., 2009)—and the cross-entropy estimation of Robinson et al. (2001), which remains the standard reference for SAM balancing under scattered and unequally reliable information. Compilation *practice*, as distinct from compilation mathematics, has its own literature: Round (2003) sets out the lessons and unresolved difficulties of building SAMs for policy analysis, and estimation manuals aimed at producers—most directly Mainar-Causápé et al. (2018), the manual behind the European Commission's own African SAM programme—document the sequence of steps a compiler follows. What these do not do, and what this paper is about, is make the steps executable and the resulting artefact checkable against them. Compilation practice for the supply–use core is codified in the SNA and its handbooks (United Nations and European Commission and IMF and OECD and World Bank, 2009; United Nations Statistics Division, 2018; Eurostat, 2008); the analytics that consume SAMs are surveyed by Miller and Blair (2009) and, for CGE databases, Löfgren et al. (2002). Operationally, most African country SAMs are produced within a small number of institutional pipelines—notably IFPRI's country programmes (Thurlow, 2021), UNU-WIDER's South Africa series, of which the benchmark used here (van Seventer and Davies, 2023) is a leading example, and the Joint Research Centre's Africa programme, which has published a 2019 South African SAM disaggregated by province, gender, and formal–informal status (El Meligi et al., 2025). South Africa's 2019 accounts have thus been rendered as a SAM at least three times, independently and for different analytical purposes. The cell-level audit reported here is against the UNU-WIDER benchmark alone, but Section 5.5 compares our result with the Commission's at the macro level—a reconciliation the field performs more rarely than the existence of three renderings would suggest.

The paper's infrastructure claims sit at the junction of several adjacent literatures, each of which has solved a piece of the problem. Large international input–output databases are compiled under documented, code-driven procedures—WIOD (Timmer et al., 2015), EXIOBASE (Stadler et al., 2018), and Eurostat's FIGARO tables (Eurostat, 2019)—and open-source tooling such as `pymrio` (Stadler, 2021) has made their *consumption* programmatic; but these systems document methodology at the level of reports, not cell-level provenance, and none is designed to let an outsider audit a single published matrix against its own record. Machine-readable statistical standards (SDMX for dissemination (SDMX Initiative, 2021); the W3C PROV model for data lineage (World Wide Web Consortium, 2013)) supply exactly the vocabulary our standards proposal needs, but have not been applied to SAM construction. And the norms of reproducible computational research—Peng (2011), Sandve et al. (2013), and for economics practice Gentzkow and Shapiro (2014)—prescribe the workflow discipline the framework implements, without the accounting-identity instrument that makes SAM auditing unusually sharp. Within economics the paper joins the transparency and replication agenda (Christensen and Miguel, 2018), with that accounting twist: a SAM's identities give the auditor a powerful instrument, because any undocumented intervention must either break an identity or leave a biproportional fingerprint.

The closest existing systems are the input–output virtual laboratories, of which the Australian Industrial Ecology Lab is the developed case (Lenzen et al., 2014): registered source "data packs", scripted assembly, and constrained balancing under KRAS, built so that many analysts can compile tables collaboratively and regenerate them. The virtual-laboratory programme establishes most of what we assume here, and we claim no priority over it. Two things differ, and they are differences of purpose rather than sophistication. A virtual laboratory is built to *construct* tables, so its provenance record is oriented to the build; our concern is the inverse operation—taking a matrix someone else published, with its own prose record, and testing whether that record reproduces it. And the laboratories serve environmental-economic MRIO accounts, where the reference artefact is the database itself, rather than the nationally published SAM whose documentation is the object of audit. What we have not found is a published, cell-level replication of a national SAM *from its own stated sources*; if such an audit exists we would welcome the correction, since the paper's argument is strengthened, not weakened, by evidence that the exercise is feasible more often than the record suggests.

2.2 Why current workflows resist reproduction

The prevailing construction workflow—official tables copied into spreadsheets, reconciled by hand, described afterwards in prose—fails five requirements at once, none of them because practitioners are careless (Table 1). *Transparency*: judgments act on individual cells but are reported, if at all, at the level of methods paragraphs. *Versioning*: a spreadsheet's history is its author's memory; source vintages and revisions are rarely pinned. *Provenance*: a finished cell does not say which source, which transformation, and which manual edit produced it. *Auditability*: a reader can check that the matrix balances, but not why any cell holds its value. *Reproducibility*: rebuilding the matrix from its stated sources is a research project, not a command—as this paper demonstrates by undertaking exactly that project. To our knowledge it is the first full cell-level replication audit of a published national SAM from primary sources; that such an audit is novel is itself evidence of the problem.

3 Data

Every input is recorded in a source registry with publisher, vintage, access date, licence status, and SHA-256 hash; raw files are never modified, and artefacts derived from them carry lineage to the registry entry. The South African sources are:

Table 1: Prevailing SAM construction workflow versus the framework used here.

Requirement	Prevailing practice	This framework
Transparency	Judgments described in prose, if at all	Classifications, formulas, and rules are machine-readable files
Versioning	Source vintages implicit; spreadsheet history unrecorded	Every input registered with vintage, access date, SHA-256 hash
Provenance	Finished cells carry no origin	Cells trace to source series, transformation, and logged adjustment
Auditability	Balance checkable; cell values not	Identities enforced as executable checks; adjustment log published
Reproducibility	Rebuild is a research project	Rebuild is a command (scripts 01–12 of the replication package)

- **Supply and use tables** (Stats SA Report 04-04-03): the 2018–2019 workbook with supply and use tables at 124 industries and 108 products, plus the accompanying methodological report. These are the sole inputs to the production and commodity blocks.
- **Benchmark SAM** (UNU-WIDER Technical Note 2023/1): the distributed workbook of van Seventer and Davies (2023)—two 209-account SAMs (occupational and educational wage detail), an account dictionary, and industry/product concordances—plus the technical note itself, from which we transcribe the 46 macro cell formulas and three mapping tables. The workbook is used *only* as the audit target and as the published source of its own concordances; no benchmark value enters our SAM except where explicitly flagged (the single unpublished input, in the as-built scenario only).
- **SARB Quarterly Bulletin** (December 2022 bulk files): all 69 annual KBP series required by the technical note's formulas. We verify the vintage directly: household consumption and imports in these files equal the benchmark's corresponding cells exactly, at the precision the bulletin files report (this vintage check is the one comparison made at source precision rather than at the R0.5 million audit tolerance of Section 4.1).
- **Annual Financial Statistics** (Stats SA P0021): the PPE schedules (asset type × SIC industry) from both the 2019 preliminary and 2019 revised editions—the comparison matters, as Section 5 shows.
- **Labour Market Dynamics** 2018 and 2019 and **Living Conditions Survey** 2014/15 microdata (Stats SA via DataFirst): used only to compute aggregated shares (occupational wage bills; household groups). Under the access conditions, no microdata are redistributed; the replication package ships the scripts and the aggregated shares.

One input to the benchmark is not public: the split of import duties from other product taxes rests on customs data that its technical note describes as "made available by Statistics South Africa as unpublished data".[3] We treat this input as a first-class object: registered in the source registry, flagged in the formula table, propagated to the three cells it touches, and priced by a public proxy (Section 5.2).

[3] van Seventer and Davies (2023, item vii). The note does not describe how the data were provided or in what form; we assert nothing beyond its own wording.

The two further applications add three registered sources. For Kenya: the IFPRI 2019 Nexus SAM (Thurlow, 2021), distributed under CC-BY-4.0 via Harvard Dataverse, and the KNBS Economic Survey 2020 (Kenya National Bureau of Statistics, 2020), whose current-price national-accounts tables supply the macro controls; the 2009 benchmark supply and use tables exist in condensed form in the Economic Survey 2014, with the full 151-product database available from KNBS on request but not published machine-readably. For the Netherlands: the Eurostat dissemination API (Eurostat, 2026), from which three requests retrieve the 2019 supply table, use table, and non-financial sector accounts (datasets `naio_10_cp15`, `naio_10_cp16`, `nasa_10_nf_tr`); the retrieved JSON responses are registered and hashed like every other raw input.

4 Method

4.1 Terms and architecture

Six related terms recur and are not interchangeable; we fix them here. *Reproducibility*: rerunning the released code on the registered inputs yields the released artefact, to a stated tolerance. *Replication*: independently rebuilding an artefact from its stated sources and methods; *reconstruction* is our implementation of it, assembling a new SAM from primary sources under disclosed rules. *Auditability*: the property that each discrepancy between an artefact and its record can be traced, bounded, or explicitly classified rather than left latent (Appendix B classifies this paper's findings on exactly that scale). *Provenance*: the recorded lineage from each cell back to its source series, transformations, and adjustments. *Deterministic reconciliation*: detecting divergence by arithmetic—an identity that fails to close, a grid search with a unique survivor—rather than by reading documentation. Unless stated otherwise, "exact" in this paper means agreement within R0.5 million, the rounding grain of the benchmark's distributed workbook; where a comparison is made at a source's own reported precision (the SARB vintage check of Section 3), we say so explicitly.

Figure 1 shows the framework's architecture: eight stages from registered sources to reproducible artefacts, with the four commitments of Section 1 enforced at the stages where they bind. The same stages serve construction and audit—auditing is construction plus a comparison target.

4.2 Pipeline

The South African pipeline is twelve construction and audit scripts (01–12) over the open toolkit; a thirteenth generates the paper's exhibits, and scripts 14–15 implement the Kenyan and Dutch applications of Section 6. The design principle is that *nothing is classified by convention.* Account codes, prose descriptions, and header labels are treated as claims to be verified, never as facts. Concretely:

1. **Ingestion with internal checks.** Readers for the supply and use tables recompute every embedded total and verify the layout's accounting identities: for each product, supply at purchasers' prices must equal supply at basic prices plus product taxes plus

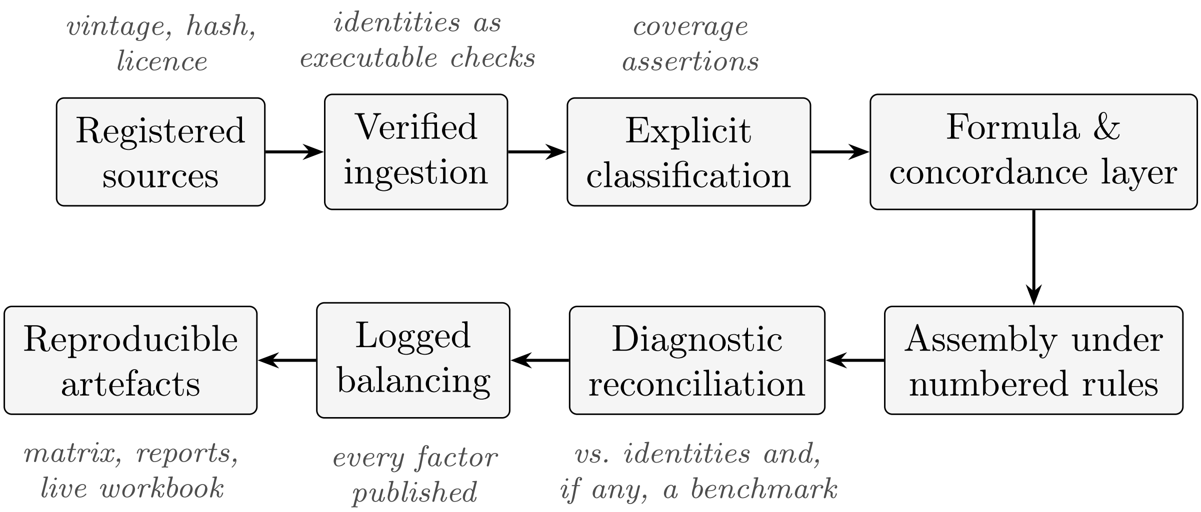

Figure 1: The framework's eight stages. Auditing and construction share the pipeline; an audit adds a comparison target at the reconciliation stage.

trade and transport margins; for each industry, output must equal intermediate inputs plus value added; and value added must equal the sum of its components—compensation of employees, other net taxes on production, and gross operating surplus or mixed income (coded $D1$, $D2/3$, and $B2/3$ in the published tables, summing to value added $B1$).[4] The official 2019 tables pass all checks at a R0.5 million tolerance.

2. **Explicit classification.** The 209 benchmark accounts are mapped to a canonical taxonomy (activity, commodity, factor, household, ...) in one reviewable table extracted from the workbook's own dictionary sheets; a coverage assertion guarantees that every account in the matrix is classified and nothing phantom is. Classification by prefix is refused by construction—for good reason, as the results show.

3. **Concordances as data.** The 124→61 industry and product mappings, the labour-survey industry table, and the asset and industry tables of the financial-statistics source are parsed from the benchmark's published materials into CSV concordances with completeness validation (every source category maps exactly once).

4. **Institutional cells as formulas.** The technical note defines each macro cell in prose as a combination of SARB Quarterly Bulletin series. We transcribe all 46 definitions into a machine-readable formula table—one row per cell, holding the signed combination of series codes—which a small evaluator computes mechanically.[5] The one unpublished input enters this table as a flagged parameter evaluated under two scenarios: *as built* (the benchmark's value) and *public proxy* (the closest published series), so that its effect is

[4] These are the SNA transaction codes used by Stats SA's published tables: $B1$ gross value added, $D1$ compensation of employees, $D2/3$ other taxes less subsidies on production, $B2/3$ gross operating surplus / mixed income (United Nations and European Commission and IMF and OECD and World Bank, 2009).

[5] For example, the cell (households ← labour) is the single series KBP6240J, "compensation of residents"; the cell (government ← production taxes) is KBP6600J − KBP6601J. Two rest-of-world cells that the note itself defines as residuals of other entries are stored as such and resolved recursively, e.g. (rest of world ← labour) = (labour income) + (labour from abroad) − (labour to households).

measured rather than absorbed.

5. **Share systems from microdata.** Capital-type shares from the AFS PPE schedules; occupational wage-bill shares from LMD 2018–2019 (weighted employment × average wage, national occupation wage for empty cells, two-year averaging); household groups as weighted expenditure deciles with the top decile in five 2 per cent bands from the LCS.

6. **Assembly under numbered rules.** The full 209-account SAM is assembled from the above; every point where the benchmark's choice is unpublished or undocumented is replaced by an explicit rule, numbered R1–R7 and reproduced in Appendix A,[6] so that the difference between our SAM and the benchmark is, by construction, the union of disclosed rules.

7. **Balancing with a log.** Residual imbalances—confined to the household income block—are closed by RAS with fixed payer totals and group outlay targets; the implementation refuses negative seeds, reports convergence, and returns the multiplicative factor of every adjusted cell, which is published in the workbook.

4.3 Audit method: reconciliation and recovery

The audit compares each constructed block with the benchmark at the finest common level and treats any discrepancy as a diagnostic object. Three patterns recur. *Identity-guided attribution*: a mismatch is traced by asking which accounting identity it violates and what movement would repair it (this is how mislabelled accounts and valuation mismatches announce themselves). *Grid recovery*: when documentation is silent over a small discrete choice set—which source edition, which measurement column, which signs—we evaluate all candidates against the benchmark and accept a choice only if it is uniquely best, reporting the margin.[7] *Scenario pricing*: where an input is genuinely non-public, we quantify the cost of its best public substitute rather than absorb it silently.

5 Results

Table 2 summarises the audit in the metrics that matter for reproducibility; the subsections unpack each row. One remark applies throughout: none of the findings below is a criticism of the benchmark's authors, whose documentation is far above the norm for the genre and whose distributed workbook is what made a cell-level audit possible at all. Small reconciling interventions are a normal and necessary part of SAM compilation; our findings concern whether the record allows them to be seen, not whether they should exist.

[6] The rules also ship as part of the replication package and are printed on a dedicated sheet of the generated workbook.

[7] Two instances matter below. For one aggregation formula, the six quoted series were each allowed a coefficient of −1, 0, or +1; of the $3^6 = 729$ combinations, exactly one reproduces the benchmark, and the runner-up misses by R2.5 billion. For the capital-stock split, the grid crossed the source's two published editions (2019 preliminary versus 2019 revised) with two candidate stock measures (opening versus closing carrying value); "recovering the vintage and measure" means identifying which edition–measure pair the benchmark used.

Table 2: Headline replication metrics: our reconstruction from publicly obtainable sources, audited against the published benchmark. Comparisons use the tolerance of Section 4.1 (R0.5 million, the benchmark workbook's rounding grain) unless stated; "comparable" cells are those nonzero in both our matrix (6,779 nonzero cells) and the benchmark's (6,737).

Block	Metric	Result
Production (61 activities)	output, intermediates, prod. taxes, VA	exact (≤R0.5m), 61/61 each
Final demand (5 categories)	per-commodity agreement	exact, all commodities (316/316)
Institutional macro (SARB)	45 computable cells	43 exact; 45 within 1%; max 0.139%
Capital split (6 types)	activity–type shares	median 0.18pp; 73.6% within 1pp
Occupation split (10)	activity–occupation shares	median 0.04pp; 88.7% within 1pp
Household groups (14)	wage-income anchor	mean 0.74pp; max 1.9pp
Full matrix (209 accounts)	6,732 comparable cells	62.5% equal within R0.5m
Balanced SAM (v0.2)	max account imbalance	R7m; RAS factors in $[0.907, 1.143]$
Non-public inputs	count; priced effect of proxy	1; mtx +16.1%, stx −1.5%
Divergences (record vs. data)	catalogued in five classes	8 (Sec. 7; App. B)

5.1 The production and commodity blocks reproduce exactly

Aggregating the official supply and use tables through the published concordances reproduces the benchmark's production side to within R0.5 million on *all* 61 activities: gross output, intermediate consumption, production taxes, and value added at factor cost each deviate by 0.0000 per cent. Final demand reproduces exactly per commodity in every category (households 93/93 commodities, government 3/3, exports 97/97, fixed capital formation 21/21, and inventories-plus-residual 102/102—316/316 in all, confirming that the benchmark's stock-change account absorbs the use table's residual column). The benchmark matrix itself is exactly balanced (maximum |row−column| gap below R0.001 million across the 210 accounts of its distributed matrix; the comparison below is on the 209 accounts of our rebuild). For the macro comparison of Section 5.2 the benchmark is aggregated to the technical note's Table-2 accounts, the margin account consolidating away exactly.

Three discrepancies had to be resolved to get there, and each is a finding about naming rather than about numbers. The account `atx` carries an activity-style prefix but is taxes on production (it equals the use table's $D2/3$ exactly on 61 of 61 activities); the accounts `cler` and `craf` carry the commodity prefix but are the clerical and craft *occupation* accounts—before reclassification they masquerade as a systematic R479 billion overstatement of intermediates spread across every activity; and the value-added comparison closes only at factor cost, since the benchmark parks production taxes outside its factor payments. The workbook's own dictionary sheet later confirmed all three identifications—and added a fourth: the descriptions of the capital accounts `mach`, `nitc`, and `trnp` are rotated by one position relative to the technical note's authoritative list.

5.2 The institutional macro block from SARB series

Evaluating the technical note's 46 macro cell formulas over the December 2022 SARB Quarterly Bulletin series reproduces 43 of the 45 computable cells exactly; all 45 are within 1 per cent and the largest deviation is 0.139 per cent. (Of the 46 cells, 43 evaluate directly from KBP-series

Table 3: Institutional macro cells requiring correction or judgment (R million; deviations against the benchmark). "Proxy" replaces the unpublished customs figure with the public series KBP4590J.

Entry	Cell (receiver ← payer)	Computed	Benchmark	Dev. %	Proxy dev. %
vi	stx ← commodities	519,805	519,805	+0.00	-1.51
vii	mtx ← commodities	48,934	48,934	+0.00	+16.09
xv	enterprises ← enterprises	857,644	857,644	+0.00	+0.00
xvii	government ← enterprises	428,816	429,412	-0.14	-0.14
xxviii	enterprises ← government	686,931	687,527	-0.09	-0.09
xxxiv	government ← stx	519,805	519,805	+0.00	-1.51
xxxv	government ← mtx	48,934	48,934	+0.00	+16.09
Remaining 39 cells: exact (43 of 45 computable within 0.01%)					

formulas and two are defined residually from the others—together the 45 "computable" cells—while the 46th depends on the unpublished customs input priced below.) Getting there required three corrections that are themselves results:

- **A garbled formula, uniquely recovered.** No signed combination of the six series quoted for the intra-enterprise cell reproduces the benchmark. Searching all 3^6 coefficient vectors in $\{-1, 0, 1\}$ yields exactly one exact solution—with different signs than the prose and one quoted series unused—and the next-best candidate misses by R2.5 billion.

- **The unpublished input, priced.** The product-tax cells subtract the benchmark's unpublished customs figure (R48,934m), not the public series the note cites (KBP4590J, R56,805m); the note itself prints both variants in different places. Under the public proxy, import duties deviate by +16.1 per cent and product taxes by −1.5 per cent—the total measured cost, in this SAM, of its one non-public input.

- **An unrecorded reconciling adjustment.** Twin gaps of exactly −R596 million appear in the two government–enterprise cells. A data-revision explanation is tested and set aside (the June and December 2022 bulletins carry identical values for every series involved). Adding +596 to both directions of a bilateral flow is the unique minimal edit that preserves both accounts' balances, so the pattern is consistent with a routine reconciling adjustment made during compilation and not recorded in the note—exactly the kind of small, legitimate intervention that cell-level provenance would turn from a discovery into a declaration.

Table 3 reports the cells where these three findings act, under both scenarios; all other computable cells are exact.

5.3 Share systems: recovery and honest residuals

Capital. The technical note specifies neither the AFS edition nor the stock measure behind its six-way capital split. Grid-testing editions × carrying-value columns leaves *2019 revised, closing values* as the only candidate in the grid consistent with the benchmark (median share deviation 0.18 percentage points; 73.6 per cent of activity–type shares within 1pp), against 0.72pp for the

Table 4: Grid recovery of the capital split's undocumented basis: agreement with the benchmark's activity–type shares by AFS edition and stock measure.

AFS edition	Stock measure	Median |diff| (pp)	Within 1pp (%)
2019 preliminary	opening	1.05	49.1
2019 preliminary	closing	0.72	57.9
2019 revised	opening	0.72	58.2
2019 revised	closing ← recovered	0.18	73.6

preliminary edition that the AFS 2019 publication page supplies by default. Residual outliers are concentrated and interpretable: posts and telecommunications deviates by up to 34pp on network equipment, consistent with a sector-specific reclassification made during compilation and not reported in the note; and six activities (agriculture, education, financial intermediation, insurance, public administration, non-observed) are absent from the AFS altogether, so their published splits rest on undocumented sources.

Table 4 shows the recovery grid.

Occupations. Implementing the note's wage-bill procedure on the LMD microdata reproduces the benchmark's 10×61 occupation shares with median deviation 0.038pp (mean 0.86pp; 88.7 per cent of 610 shares within 1pp; maximum 24pp in two small-sample industries). A methodological aside earns its keep here: our own first pass silently dropped the private-households industry—6,202 observations and the home of the domestic-worker occupation—because a parser assumed three-digit industry codes. The pipeline's coverage assertion caught it; conventions failed symmetrically on both sides of the audit.

Households. Two dictionary-free anchors test the benchmark's group definition, and they disagree—informatively. The wage-income anchor identifies *household-level* expenditure deciles (top decile in five 2 per cent bands): under that definition the LCS wage-income distribution matches the benchmark's labour-to-household block at mean 0.74pp (max 1.9pp), while per-capita grouping fits markedly worse (mean 1.8pp, max 7.9pp). The consumption anchor cannot corroborate it: it fits poorly under the household-level grouping (up to 5.9pp) and in fact sits closer to the per-capita alternative (mean 0.71pp against 1.46pp). The note itself explains why the consumption block is uninformative here: its consumption shares were carried over from an earlier SAM rather than recomputed from the LCS—inherited provenance that a fresh, documented build cannot and should not reproduce, and that decouples the consumption pattern from whichever grouping the income side uses. The identification therefore rests on the wage anchor alone.

Table 5 details the occupation system by occupation, and Figure 2 plots the full distribution of share deviations for the two microdata-based systems: the mass of both distributions sits far below one percentage point, with thin, interpretable tails.

5.4 Assembly and balancing

Assembling the full 209-account SAM from our registered inputs under seven numbered rules yields 6,779 nonzero cells. Of these, 6,732 are *comparable*—nonzero in both our matrix and the benchmark's (which has 6,737)—and 62.5 per cent of the comparable cells equal the benchmark

Table 5: Occupation shares versus the benchmark, by occupation (deviations in percentage points across 61 activities).

Occupation	Median |diff| (pp)	Max (pp)	Within 1pp (of 61)
Managers	0.340	19.02	42
Professionals	0.068	24.40	50
Technicians	0.065	16.13	53
Clerks	0.080	14.94	53
Service and sales	0.028	10.01	54
Skilled agriculture	0.000	0.84	61
Craft	0.061	17.26	55
Plant/machine operators	0.063	10.27	55
Elementary	0.087	22.81	58
Domestic workers	0.000	1.28	60

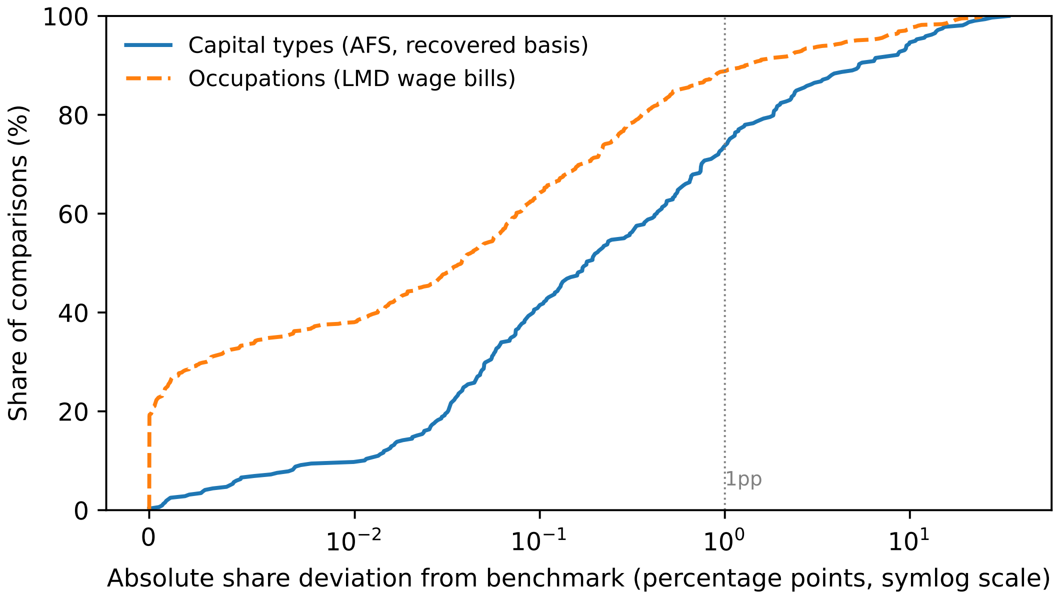

Figure 2: Cumulative distribution of absolute share deviations from the benchmark for the capital-type and occupation systems.

within the R0.5 million tolerance: the entire production and commodity side. Every material imbalance lies in the 14 household groups (maximum R100 billion, 2.0 per cent of value added at the ninth decile): precisely the footprint of the proxy distribution rules, since group-specific income composition is the one place our obtainable inputs are coarser than the benchmark's item-level survey work. Balancing the household income block by RAS—payer totals and group outlays held fixed as documented targets— closes the household imbalance to R0.06 million with adjustment factors confined to $[0.907, 1.143]$, leaves every other account within R7 million— under 0.0002 per cent of GDP, and the sense in which "balanced" is used throughout—and moves the block slightly *toward* the benchmark's item-based values. The result, SAM v0.2, ships as a workbook whose totals and balance checks are live formulas and whose 266 balancing factors are printed on their own sheet.

Table 6: The 2019 South African SAM of this paper against the independently constructed SAM of El Meligi et al. (2025), on the macro aggregates both report.

Aggregate (R billion)	This paper	JRC	Dev. %
Value added (factors from activities)	4,939	4,955	+0.3
Intermediate demand	5,992	5,989	−0.0
Household consumption	3,589	3,605	+0.5
Factor income to households	3,272	3,300	+0.8
Government consumption	1,104	1,101	−0.3
Investment and stock change	890	890	+0.0
Exports	1,502	1,502	−0.0
Imports	1,532	1,530	−0.1

5.5 External corroboration: an independently constructed SAM

Every comparison so far is against the benchmark this paper set out to audit, which leaves one question open: whether the reconstruction is right, or merely faithful to the artefact it was built to reproduce. A second, independently constructed SAM for the same country-year provides the first check that is not of our own making. The Joint Research Centre published one in November 2025 (El Meligi et al., 2025): 449 accounts against our 209, disaggregated by province, gender, and formal–informal status, estimated by entropy methods rather than assembled deterministically, and distributed openly under the Commission's reuse licence. It was not consulted during construction; the comparison below was made after this paper's SAM was final.

Table 6 sets the two side by side on the macro aggregates both report. They agree to within 0.8 per cent on every one, and to within 0.1 per cent on four of the eight, including both trade aggregates and total intermediate demand.

Two qualifications keep this in proportion. The agreement is between two *constructions*, not two measurements: both ultimately rest on Statistics South Africa's national accounts, so common inputs explain part of it, and the exercise cannot detect an error the source itself contains. And it is an agreement of aggregates: the two matrices classify at different resolutions, so a cell-level comparison would require a concordance neither publishes. What the table does establish is that two teams, working from the same official record with different methods and different disaggregation, arrive at the same macro picture—and that our reconstruction's departures from the benchmark, catalogued above, are not artefacts of our own pipeline.

The JRC's file also earns a point the audit method is well placed to make. It balances *exactly*: all 449 accounts close to within 10^{-9} at the precision distributed, which is better than the Kenyan file of Section 6.1, where rounding unbalances the published matrix, and better than the Dutch matrix this paper distributes. Distributing at full precision is a small discipline with a large effect on what a later auditor can do, and it is worth recording when a producer gets it right.

The same distribution also carries two divergences between record and artefact of the kind catalogued in Section 7, both apparent from reading its accompanying files against its contents. Its value column is headed *Million ZAR* while the values are expressed in billions, as a comparison with published national accounts indicates (value added of 4,955 against a 2019 GDP of approximately 5,078 billion rand). The readme distributed inside the archive describes the dataset as the "social accounting matrix for Nigeria 2019", the title of the same programme's Nigerian SAM. Neither affects a number in the matrix, and neither is likely to mislead a careful user. Both are of the type that an ingestion-time check on a machine-readable record identifies before release, which is the case this paper makes for such checks generally.

6 Two further applications: the replicability spectrum

The South African audit is the framework's deep case, but a method whose value is generality must be shown doing different jobs under different data regimes. This section applies it twice more, at the two ends of the public data spectrum. In Kenya, where the machine-readable supply–use detail behind the country's leading SAM is not published, the framework establishes what *can* be verified from public aggregates—and shows exactly where verification must stop. In the Netherlands, where complete harmonised tables are a single API call away, the framework runs in the opposite direction: not auditing an existing SAM but *generating* one, with no benchmark consulted at any point. Together with South Africa the three cases trace a replicability spectrum that we argue is the right way to think about SAM infrastructure worldwide.

6.1 Kenya: auditing at the aggregate boundary

The most widely used Kenyan SAM is IFPRI's 2019 Nexus matrix (Thurlow, 2021), distributed as an open CC-BY dataset. Its construction, like that of the Nexus programme generally, reconciles scattered sources by cross-entropy estimation (Robinson et al., 2001), so cell-level replication of the kind achieved for South Africa is not the relevant test: a CE-reconciled matrix is a model output, and the honest question is what its public record allows an outsider to verify. The framework's answer has three layers.

Structure verifies. The distributed file parses into 108 accounts whose groups (42 activities, 42 commodities, five factors, ten representative households, and the transaction-cost, enterprise, government, tax, accumulation, and external accounts) match the data paper's stated design exactly. The matrix balances only to the file's own integer rounding—values are printed in whole Sh billions, and account imbalances reach Sh 5 billion—which is itself a small finding about distribution practice: rounding a balanced matrix unbalances it.

Macro controls verify approximately, and the deviations are informative. Table 7 compares the SAM's implied national aggregates with the published current-price tables of the KNBS Economic Survey 2020. The production side sits a uniform 1.0 per cent below the published figures—GVA, product taxes, and GDP all shifted together—which is the signature of a vintage difference in the then-provisional 2019 accounts, not of a methodological choice (Section 7's class 3). Fixed capital formation, inventories, and imports agree essentially exactly. The large deviations are concentrated in private consumption (−9.2 per cent), government consumption (+9.1 per cent), and exports (+13.6 per cent), and they have a structural explanation: KNBS's own expenditure table carries a production-versus-expenditure discrepancy of Sh −352 billion (3.6 per cent of GDP), and a balanced SAM cannot carry a statistical discrepancy—the recon-

Table 7: Kenya: the IFPRI 2019 Nexus SAM against published KNBS controls (Sh billion, current prices; Economic Survey 2020, Tables 2.1 and 2.7, 2019 provisional).

Aggregate	SAM	KNBS	Dev. %
GVA (all activities)	8,825	8,912	-1.0
Taxes on products	821	829	-1.0
GDP at market prices	9,646	9,740	-1.0
Private consumption (incl. NPISH)	7,295	8,035	-9.2
Government consumption	1,387	1,271	+9.1
Gross fixed capital formation	1,631	1,632	-0.1
Changes in inventories	63	63	-0.2
Exports of goods and services	1,331	1,172	+13.6
Imports of goods and services	2,071	2,082	-0.5
Statistical discrepancy	0	-352	—

ciliation *must* allocate it somewhere. The deviation pattern is consistent with the discrepancy, together with associated source and definitional differences, being absorbed primarily through the consumption and trade aggregates; aggregate comparisons cannot identify the allocation uniquely, and that very limit is the point. Nothing in the public record documents the allocation at the level where it acts, which is precisely the provenance gap the paper's proposal addresses.

Cell-level audit is bounded by publication practice, not by method. The SAM's supply–use core descends from KNBS's 2009 benchmark tables, whose full balanced database (151 products by 81 industries) is, per the Economic Survey, available from KNBS on request but published only as condensed print tables. We did not request it, and the boundary tested here is therefore a specific one: what an auditor can verify from the public record alone, without privileged access or correspondence with the producer. That is the boundary a reader of a published SAM faces, and the one the paper's standards proposal addresses; an audit conducted on requested files would test a different and more permissive question. Two consequences follow. The Kenyan evidence below is confined to structure, grain, and macro controls—the depth the public record supports—and the claim that the South African pipeline would extend to the Kenyan detail remains untested: the reader for the KNBS layout, and its concordances, would still have to be written, even though the method behind them is unchanged. That boundary—audit feasible in principle, unreachable from what is published—defines the middle of the replicability spectrum.

6.2 The Netherlands: benchmark-free generation

The third application inverts the exercise. For a country inside the European statistical system, the framework should need no benchmark, no technical note, and no manual acquisition: the ESA transmission programme puts harmonised supply–use tables and sector accounts on a public API (Eurostat, 2008). We test that claim on the Netherlands for 2019. Three API calls retrieve the supply table, the use table, and the full sequence of non-financial sector accounts (about 180 KB in total); a 70-line JSON-stat reader added to the toolkit decodes them, and a single script assembles a macro accounting matrix and balances it into a macro SAM. No published SAM was consulted at any point during construction or validation—though, as the results below make clear, that is a statement about SAMs and not a claim of independent verification: the checks available here are internal consistency and agreement with published national-accounts aggregates drawn from the same compilation system that supplies the inputs.

Two results follow. First, *the official tables are internally exact*: every identity the toolkit enforces—output and value-added identities for all 65 industries, the import identity, and the supply–demand balance of all 65 products—closes at a EUR 1 million tolerance with zero findings. The contrast with the audit cases is the point: when a statistical office publishes machine-readable tables under a harmonised standard, deterministic verification is free. Even here, however, the framework's refusal to classify by convention earned its keep: Eurostat's category lists mix aggregates with overlapping sub-details (NACE section codes alongside their own divisions), and a naïve selection double-counts; the toolkit's coverage-based partition, guarded by an assertion that the chosen codes tile the total exactly, caught both this and a code collision between the NACE section P (education) and the ESA transaction P-codes. Names mislead in Luxembourg as in Pretoria.

Second, *the generated matrix closes to within 1.5 per cent of GDP with every residual attributed—and a logged balancing step then closes it exactly.* We are careful with terminology here: a SAM is defined by every account balancing, so the direct output of the generation—the 18-account *macro accounting matrix*—is the diagnostic product, and the SAM is what balancing makes of it. The matrix routes factor income, taxes, transfers, and the ESA bridging items (the pension-fund adjustment D8, capital transfers D9, net lending B9, and the residents-abroad consumption adjustments) through explicit instrument accounts; GDP at basic prices emerges at EUR 724,960 million, on the published figure, and compensation of employees closes to the euro across four series. Both agreements are internal to the Dutch national accounts: the four series and the published aggregate are outputs of one compilation system, so what they establish is that our assembly preserves CBS's own consistency, not that it is independently corroborated. Table 8 reports the residual of every account before balancing: twelve are exactly zero, and the six that are not are all below 1.5 per cent of GDP and attach to known ESA boundary items (the domestic-versus-national concept in compensation, mixed-income attribution, capital-account detail). Consistent with the framework's first commitment, the residuals are published, not hidden; the balanced macro SAM is then produced by the toolkit's logged RAS step: it converges to a maximum residual below EUR 0.01 million at full precision, with all 70 adjustment factors inside $[0.9832, 1.0282]$, concentrated on the rest-of-world cells—exactly where the attributed residuals sat—and every factor ships with the matrix. One caveat applies to the distributed artefact, and it is the same one this paper notes for the Kenyan file: the CSV prints values at EUR 0.1 million precision, so recomputing account residuals from the distributed file yields rounding imbalances of up to EUR 0.2 million; the full-precision matrix regenerates from the replication package. The balancing choices (the treatment of two negative cells, the target vector, and the choice of RAS), the invariants they preserve, and a target-vector sensitivity check are documented in Appendix C.

The cost ledger deserves its own sentence, because it carries the paper's economic claim. The South African reconstruction required four public providers, a registered microdata access, and bulk downloads diagnosed around a broken online query system; the Kenyan case required locating condensed print tables inside a 322-page yearbook; the Dutch case required three API calls and a single working session, and the 70-line reader that enabled it works unchanged for

Table 8: Netherlands: account residuals of the generated 2019 macro accounting matrix, before balancing (no benchmark consulted; GDP at basic prices EUR 724,960m). The logged RAS step closes all residuals with adjustment factors in $[0.9832, 1.0282]$.

Account	Residual (EUR m)	% of GDP
Corporations	+1,532	+0.21
Government	+1,443	+0.20
Households	+9,111	+1.26
Labour	-4,193	-0.58
Rest of world	-10,830	-1.49
Savings–investment	+2,937	+0.41
Remaining 12 accounts (production, taxes, capital, instruments): residual zero		

every country in the ESA transmission programme. Out-of-sample runs for Austria and Poland confirm that the generator transfers—and that confidentiality suppression remains a cell-level constraint even under harmonised publication; Appendix D reports the results. Conditional on the toolkit, the marginal implementation effort for a verified macro SAM is therefore shaped largely by the accessibility and structure of the published data—the replicability spectrum restated as effort.

7 Discussion

7.1 A taxonomy of divergence between record and artefact

Set side by side, the audit's findings sort into five classes, and it is worth stating plainly that every reported finding was *first detected* by arithmetic—an identity refusing to close, a grid search returning a unique survivor—and only then interpreted against the source record (Appendix B states each finding's evidentiary status):

1. **Names mislead.** Account codes mislabel their contents (`atx`, `cler`, `craf`); dictionary descriptions rotate against their codes (`mach`/`nitc`/`trnp`). Any pipeline that classifies by prefix or label inherits these slips silently; ours refuses to, and both of the audit's largest apparent anomalies (R479 billion of phantom intermediates; a 125 per cent value-added "error") dissolved into classification fixes.

2. **Prose diverges from data.** A formula's printed signs do not reproduce the matrix they describe; the same cell is printed with two different values in two places. The data adjudicate.

3. **Vintages drift.** The default download of a source (AFS 2019 preliminary) is not the edition actually used (2019 revised); nothing in the record says so, and only a grid test can tell.

4. **Inputs are unavailable.** One input is unpublished; its effect (+16.1 per cent on import duties under the public proxy) is invisible until priced.

5. **Adjustments go unrecorded.** A symmetric R596 million reconciling edit balances two accounts and appears nowhere in the record; inherited consumption shares import a different survey vintage's processing into a matrix whose documentation implies otherwise.

To repeat the point made in Section 5: these are properties of the *medium*—prose describing spreadsheets—not of the authors, whose record-keeping is far better than the norm. That is precisely what makes the count suggestive: if the best-documented SAM in its class carries five classes of silent divergence between record and artefact, there is little reason to expect fewer in matrices published with no dictionary, no formula note, and no distributed cell-level file—and for those, the audit reported here could not even be attempted, because its first step is a distributed matrix to compare against. One deep audit cannot establish a distribution over the population of SAMs, and we do not claim it does; what it establishes is that the divergences are not hypothetical, and that detecting them required a machine-readable record that most published SAMs do not supply.

The two further applications confirm that the taxonomy travels. Kenya exhibits class 3 (the SAM's production side sits a uniform 1.0 per cent from the Economic Survey print, the signature of a provisional-accounts vintage) and class 5 at national scale (the accounts' own 3.6-per-cent-of-GDP discrepancy is absorbed by the reconciliation with no cell-level record of where). The Netherlands, with the cleanest data of the three, still exhibits class 1: Eurostat's mixed-granularity category lists and a NACE/ESA code collision would silently double-count under prefix-based selection, and were caught only by the toolkit's tiling assertion.

7.2 From audit to construction workflow

The framework is not only a microscope; it is a factory. The same pipeline that audits the benchmark *builds* a SAM: scripts ingest and verify the official tables, assemble the production and commodity blocks mechanically, compute the institutional cells from published series, apply survey-based share systems, and balance the result with a logged adjustment. What remains for the economist is exactly the residue that identities and obtainable data cannot determine—in this application, the customs split, six sectors' capital shares, and the distributional detail of household income. That inversion is the practical payoff: expert attention stops being spread across ten thousand spreadsheet cells and concentrates on the handful of judgments that genuinely require it, each of which the framework forces into the open as a numbered rule. Updating the matrix for a new benchmark year, or transferring it to a country with comparable sources, reuses everything except those judgments.

7.3 Implications: a standards proposal

Two practices would have made most of this paper's detective work unnecessary, and both are cheap. We state them as a proposal for producers of SAMs—statistical offices, research institutes, and international programmes alike:

1. **Publish cell-level provenance.** Each cell of a released SAM should carry its source series, transformation, and any manual adjustment. The R596 million edit and the inherited

consumption shares would then be declarations rather than discoveries, and reconciling interventions—a normal part of compilation—would be visible instead of latent.

2. **Publish machine-readable construction rules.** The benchmark's technical note was nearly a program; had it been an actual one (as our transcribed formula table now is), the sign discrepancy could not have survived contact with the data, and replication would be a command rather than a project. Concordances, account dictionaries, and balancing targets belong in the same release.

Neither requires changing a single methodological choice, and both are by-products of building the matrix in code to begin with. Our toolkit is offered as working infrastructure for exactly that: an open reference implementation of registered sources, executable identities, explicit classification, formula tables, diagnostic balancing, and workbook generation, intended to be reused wholesale for other countries and years.

7.4 Cost: a three-country ledger

We report the development ledger factually and claim nothing causal about labour saving from it. The South African case required registering sources from four providers, one registered microdata access, and bulk downloads worked around a non-functioning online query system; the Kenyan case required locating condensed print tables inside a 322-page yearbook; the Dutch case required three API calls and roughly 300 new lines of code (70 of them a reader that now serves every ESA-transmitting country). The claim we do make is conditional and testable: the marginal cost of the *next* SAM built on the toolkit is largely the cost of its country-specific concordances and share systems, and the Dutch application is a first operational indication of that claim at the macro level. Across the three cases, publication format emerges as a major determinant of marginal implementation effort—which is what the standards proposal above is for.

7.5 Limitations

The deep audit inherits its benchmark's scope—one country, one year, one SAM family—and the two further applications, while they demonstrate transfer, are deliberately shallower: Kenya stops at structure and macro controls because the machine-readable detail is not public, and the Dutch exercise generates and balances a macro SAM, not a full multi-sector SAM with distributional detail (that extension is mechanical for the production core, which the API supplies at 65 industries, but household disaggregation would require the national budget-survey work the South African case illustrates). Six activities' capital splits and the item-level detail of the household systems remain beyond the inputs we hold, and our balanced household block is a disciplined proxy, not a reconstruction— closing that gap requires the survey's item-level classifications, which are obtainable and queued. The recovered choices (editions, measures, signs) are unique within their tested grids, not proofs of intent; where no grid candidate fits exactly—as in the capital split—the true basis may lie outside the set we enumerated.

Two limits are structural rather than circumstantial, and neither is removed by better data. The first concerns what the method is able to detect. Every finding reported here was identified because it left an arithmetic trace: an identity that did not close, a grid search returning a single survivor, or a pair of gaps whose symmetry was unlikely to be coincidental. Divergences that leave no such trace pass through this framework without being detected. These include compensating errors, adjustments that happen to land on a plausible value, and errors shared between the artefact and the sources against which it is checked. The audit therefore establishes that a matrix is reproducible from its stated record, which is a narrower property than correctness. The comparison in Section 5.5 narrows this gap without closing it: an independently constructed SAM for the same country-year agrees with ours on every macro aggregate to within 0.8 per cent, which is meaningful corroboration, but it remains agreement between two constructions resting on a shared official source rather than a second measurement of the same economy. An error contained in the source record itself would not be detected by any check reported here.

The second limit is that the framework records without evaluating. It reports every adjustment factor, coverage share, and residual, but it does not encode a threshold above which a matrix should be rejected, and we do not propose one. The tolerable magnitude of a balancing adjustment depends on the account, the country, and the intended use of the SAM, so a threshold calibrated to three cases would carry more apparent authority than the evidence supports. What the framework changes is the availability of the question rather than its answer: because the factors and residuals are published, a reader can apply whichever threshold their application requires. Rejection criteria for the field would need to be agreed among producers rather than set by a single implementation.

Finally, our own pipeline's mistakes—one of which we document—were caught by the same assertions we advocate; we take that as the strongest available evidence for building them in.

8 Conclusion

A national social accounting matrix can be rebuilt from publicly obtainable sources—in this application, Statistics South Africa's supply and use tables and Annual Financial Statistics, the South African Reserve Bank's Quarterly Bulletin series, and the Labour Market Dynamics and Living Conditions surveys—with its production and commodity structure exact, its institutional macro structure exact up to one priced unpublished input and one discrepancy consistent with an unrecorded adjustment, and its distributional detail approximated within disclosed limits and closed by biproportional (RAS) balancing whose every factor is logged. The exercise yields a balanced, fully regenerable SAM; a catalogue of divergences between record and artefact, each first detected by deterministic reconciliation and then interpreted against the source record; and an open toolkit offered as reproducible infrastructure for official SAM construction. The same framework, unchanged, audits Kenya's leading SAM to the boundary its publication format imposes—verifying structure and macro controls, with a deviation pattern consistent with the accounts' own statistical discrepancy being absorbed through consumption and trade—and generates a Dutch macro accounting matrix from three API calls with no benchmark at all, every residual attributed, then balances it into a SAM with every adjustment factor logged. The three cases are one argument: SAM construction can move from an opaque artisanal exercise to an auditable, executable statistical-production process, and what separates an exactly replicable SAM from an unauditable one is not method, geography, or income level, but whether the underlying tables are published machine-readably and whether construction choices are recorded where they act. The idea worth carrying away is therefore the simple one the evidence keeps confirming: cell-level provenance and machine-readable construction rules, published alongside every SAM, would make such audits—and such SAMs—routine. The immediate next steps are the survey item-level household systems, cross-entropy balancing as an alternative adjustment principle, and extending the Dutch generation to a full multi-sector SAM across the ESA-transmitting countries.

Reproducibility and AI disclosure

The replication package comprises fifteen scripts over an open toolkit (`edikit`, released under the Apache-2.0 licence)—the South African construction and audit pipeline, the exhibit generator, and the Kenyan and Dutch applications—plus a source registry with SHA-256 hashes for every input, derived concordances and formula tables, per-script validation reports, and a workbook whose totals and balance checks are live formulas. All results in this paper regenerate from the registered sources. Survey microdata are used under DataFirst access conditions and are not redistributed—only aggregated shares appear in any output. The package is openly available at `https://github.com/kodzovee9/tekmeris` and archived at doi:10.5281/zenodo.21419195; Appendix E maps each of the paper's claims to the script that produces it and the validation report that records it.

The toolkit and the computations reported here were prepared with extensive use of a large language model (Anthropic's Claude) for code generation, document parsing, and diagnosis. No AI-generated number enters any result without passing the pipeline's accounting identities and coverage assertions; all generated code, mappings, and outputs were reviewed by the author, who assumes full responsibility for the results. The paper reports one instance (Section 5.3) where those assertions caught an AI-introduced error—which we regard as the safeguards working as designed.

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A Disclosed construction rules

Table 9 reproduces the numbered rules under which our SAM departs from the benchmark's (unpublished or unrecorded) choices. Each rule is implemented in the replication package and printed in the generated workbook; "measured consequence" refers to the comparison against the benchmark reported in Section 5.

Table 9: Rules R1–R7: explicit substitutes for non-public or unrecorded choices.

Rule	Construction	Replaces / measured consequence
R1	Import duties by commodity: total from the public series KBP4590J, allocated by import shares	Unpublished customs data; +16.1% on import duties
R2	Other product taxes by commodity: supply-table tax column minus R1	Same dependency; −1.5% on product taxes
R3	Capital-type shares for six activities absent from the AFS: economy-wide average shares	Undocumented benchmark source for those sectors
R4	Household consumption pattern: each commodity split across groups by total-expenditure shares	Benchmark's inherited item-level shares; drives most residual household imbalance
R5	Occupation-to-household wages: aggregate LCS wage-income group shares for every occupation	Benchmark's education-based expansion
R6	Other household-facing cells: LCS wage-income group shares as income-side proxy	Benchmark's item-specific LCS systems
R7	Capital income to institutions uniform across the six capital types	The benchmark's own documented rule (adopted)

B Evidentiary status of the audit findings

Not every divergence between record and artefact has the same epistemic status, and the audit methodology requires saying which is which. Table 10 classifies each South African finding by its detection mechanism and the strength of the resulting inference, on a five-level scale: *exactly recovered* (the data admit one answer and it reproduces the benchmark); *unique within tested grid* (one candidate among an enumerated finite set fits, with the runner-up margin reported); *consistent with* (the pattern matches a hypothesised intervention that is not uniquely identified); *documented* (the source record itself states the choice); and *unresolved* (the inputs needed to decide are unavailable). The table has ten rows in total: seven covering the eight record–artefact divergences catalogued in Section 7 (the two customs-related divergences—the double-printed value and the non-public input—combined in one row), plus three supplementary items reported for completeness: a recovered grouping choice, an unresolved data gap, and a residual outlier.

Finding	Detection mechanism	Evidentiary status
`atx`, `cler`, `craf` misnamed	identity match on all 61 activities	exactly recovered
`mach`/`nitc`/`trnp` rotation	dictionary cross-check against technical note	documented
Formula signs (intra-enterprise cell)	exhaustive 3^6 grid; runner-up misses by R2.5bn	unique within tested grid
Capital-split vintage and measure	edition $\times$ measure grid (0.18pp vs 0.72pp)	unique within tested grid
Household grouping variable	wage-income anchor (the consumption anchor is uninformative: inherited shares)	unique within tested definitions
R596m government–enterprise edit	balance-preserving minimal-edit pattern; vintage test refuted	consistent with
Customs input (`mtx`/`stx`)	note's own statement; priced by public proxy	documented; unresolved input
Six sectors' capital shares	absent from AFS source	unresolved
Household consumption shares	technical-note statement	documented (inherited system)
Posts/telecoms capital outlier	residual after recovered basis	consistent with (unreported reclassification)

Table 10: South African audit findings by detection mechanism and evidentiary status.

C Balancing the generated Dutch matrix: rules, invariants, and sensitivity

The balancing step of Section 6.2 involves three choices, none mechanically inevitable; this note states each rule, the invariant it preserves, and the measured sensitivity to the one genuinely discretionary choice.

Sign normalisation. The generated matrix contains two negative cells: net property-type capital income vis-à-vis the rest of the world (EUR $-10{,}286$m at (row $\leftarrow$ capital)) and the D8 pension adjustment vis-à-vis the rest of the world (EUR -168m). Each is flipped to its positive transpose before balancing: a payment of $-v$ from account a to account b is identically a payment of $+v$ from b to a. The flip preserves, cell by cell, every account's receipts-minus-payments residual and every institution's net position; it changes only the gross flows' side of the ledger, and both flips have the natural economic reading (the Netherlands is a net *recipient* of capital income from abroad). RAS requires a non-negative seed, so the alternative would be GRAS (Junius and Oosterhaven, 2003); normalisation is preferred *at this scale* because it makes the negativity explicit and reviewable, two cells being small enough to inspect individually. The preference does not generalise: a full SAM carries structural negatives throughout (subsidies, negative saving, inventory changes), and transposing each one would rearrange the matrix away from the conventional layout. For that case GRAS is the appropriate instrument, and the toolkit implements it.

Target vector. RAS needs one target per account. The generated matrix supplies two candidates—each account's receipts and its payments, which differ by exactly the attributed residual—and the rule takes their arithmetic mean: the symmetric choice that favours neither side of any account and whose row- and column-target totals agree by construction. The two one-sided choices (all-receipts, all-payments) are the extreme brackets of this decision.

Sensitivity. Rebalancing against each extreme bracket (scaled to a common total so variants are comparable cell by cell) moves no cell of the balanced matrix by more than EUR 6.4 billion—0.9 per cent of GDP, the same order as the pre-balancing residuals themselves—and no cell above EUR 100m by more than 6.6 per cent of its value. That is the expected behaviour: the target choice redistributes the attributed residuals between the two sides of each unbalanced account; it does not amplify them, and the mean-target SAM lies between the brackets. Sensitivity is regenerated by the replication package alongside the matrix.

Choice of RAS. After normalisation the seed is non-negative, so GRAS is unnecessary rather than rejected. RAS is preferred to cross-entropy estimation here because its per-cell multiplicative factors are individually interpretable and are published with the matrix—the framework's logging commitment—while cross-entropy with uninformative priors would deliver essentially the same adjustment with less transparent diagnostics (Robinson et al., 2001).

D Transfer validation across additional ESA publication regimes

After the Dutch case was complete, the packaged generator (`toolkit/recipes/build/` in the replication package) was rerun, unchanged, on two further countries chosen to bracket the harmonised system's publication practice; validation reports for both ship with the replication package.

Austria 2019 behaves like the Netherlands: 65 industries and products, zero identity findings at the EUR 1m tolerance, full commodity closure, and every pre-balancing account residual below 0.6 per cent of GDP.

Poland 2019 exposed a publication-practice boundary *inside* the harmonised system: Polish tables withhold confidential cells, so the published industry-and-product detail covers only 96.7 per cent of gross output (government final consumption detail: 92.6 per cent). The toolkit detects the suppression through its tiling and coverage assertions, anchors the macro accounts to the published totals—which remain complete and internally consistent—and reports the coverage of every affected aggregate; the generated Polish matrix closes with residuals below 1.1 per cent of GDP. Even within the ESA transmission programme, then, the replicability spectrum persists at the cell level: harmonisation fixes the format, not the disclosure.

E The toolkit and replication package: a reader's guide

The toolkit (`edikit`, Apache-2.0; repository `https://github.com/kodzovee9/tekmeris`, archived at doi:10.5281/zenodo.21419195) is a small Python library in four modules—`data` (readers for supply–use workbooks, SAM matrices, central-bank series, Eurostat JSON-stat, and concordances, each enforcing its source's accounting identities on ingestion), `model` (the account taxonomy with coverage assertions; RAS balancing with mandatory per-cell factor diagnostics), `report` (generation of workbooks whose totals and balance checks are live formulas), and `pipeline` (end-to-end generation and audit routines)—plus a test suite in which the accounting identities are the test cases. Users enter through three documented *recipes*, matching the publication regimes of this paper: `replicate/` (regenerate every number here), `audit/` (audit a published SAM: structure, then macro controls against published national accounts), and `build/` (generate and balance a macro SAM for any ESA-transmitting country). The latter two are each one command, e.g.:

```
python build_sam.py --country AT --year 2019
python audit_sam.py --sam <matrix.csv> --kinds <kinds.csv> \
                    --controls <controls.csv>
```

Step-by-step instructions live in the package's README files, which can evolve with the code; this appendix instead gives the map a verifying reader needs: which script produces each of the paper's claims, and which validation report records it (Table 11). Every script writes its report to `outputs/`; the reports, not the console, are the record.

Claim	Script(s)	Validation report
SUT identities pass; production and commodity blocks exact	01–03	`supply_/use_validation`
Institutional macro cells from SARB formulas; grid-recovered signs; priced customs proxy	04–06	`macro_sam_construction`
Capital split: vintage/measure recovery grid	07	`capital_split`
Occupation shares from LMD wage bills	08	`occupation_split`
Household grouping identification	09	`household_split`
Full assembly under rules R1–R7; comparable-cell counts	10	`assembly_report`
RAS balancing, factors logged; workbook	11–12	`balance_report`; workbook
Paper exhibits regenerate from outputs	13	`paper/tables`, `figures`
Kenya: structure, rounding grain, KNBS macro controls	14	`kenya_structural_audit`
Netherlands: generation, residual attribution, balancing and its sensitivity (Appendix C)	15	`netherlands_generation`
Austria/Poland transfer validation (Appendix D)	build recipe	`{AT,PL}_2019_generation`

Table 11: Map from the paper's claims to the scripts that produce them and the validation reports that record them. Scripts 01–12 need only the registered sources; 08–09 additionally require the DataFirst microdata (Section 3).